5.7 The Complex Exponential as a Map

WORKSHEET

The question

STEP II 1997, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What are the modulus and argument of $z = e^u(\cos v + \mathrm{i}\sin v)$ in terms of u and v?

STAGE 1 Solving $e^w = \mathrm{i}e$

Covers: finding w when $z = \mathrm{i}e$

Ask yourself What are the modulus and argument of $z = e^u(\cos v + \mathrm{i}\sin v)$, in terms of u and v?

Pause and try What are the modulus and argument of $\mathrm{i}e$? How many solutions are there?

STAGE 2 The images of grid lines

Covers: the images of the lines $u =$ constant and $v =$ constant

Ask yourself On a line $u = c$, what stays fixed and what varies? And on a line $v = d$?

STAGE 3 One-to-one pieces

Covers: the two pairs of subsets

Ask yourself Which points w map into the first quadrant? Why is that set too big to be in one-to-one correspondence with the quadrant?

Pause and try For $0 < |z| < 1$: which values of u do you need? Every argument must be hit exactly once, so how long should the interval of v be, and which ends should it include?

What's really going on

The complex logarithm

On the real line, ln is the inverse of exp. In the complex plane, $e^w = z$ has infinitely many solutions w for each $z \neq 0$, differing by multiples of $2\pi\mathrm{i}$, so "$\log z$" is not a single number. To get a genuine function you must choose a *branch*, which means choosing a strip like W_3 or W_4: the *principal branch* takes $-\pi < \arg z \leq \pi$. Every branch has a jump somewhere, along a ray from the origin called a *branch cut*. Bernhard Riemann's resolution, in his thesis of 1851, was to stop insisting that the logarithm lives on a plane: it lives on an infinite spiral staircase, a *Riemann surface*, on which it is single-valued and smooth. Riemann surfaces are now central to geometry, number theory and string theory.

Angles are preserved

Look again at Figure 5.7.1 (in the full section online). The vertical and horizontal grid lines meet at right angles, and so do the circles and rays they become. This is not a coincidence: wherever the derivative of a complex function is non-zero, the function preserves the angles between curves. Such maps are called *conformal*. Locally, a conformal map is just a rotation and a scaling, which is exactly what multiplication by the complex number $f'(z)$ does. Conformal maps are used to solve problems in fluid flow, electrostatics and heat conduction, by transforming a complicated region into a simple one where the answer is known.

Mercator's map

Here is a surprise connecting this section with Section 4.8. Gerardus Mercator's map of 1569 was designed so that a course of constant compass bearing is a straight line, which requires preserving angles. If you first project the globe stereographically onto a plane (from the north pole), then take the complex logarithm, the result is exactly Mercator's projection: meridians become vertical lines and lines of latitude become horizontal lines, just as rays and circles become horizontal and vertical lines under log. The exponential map in this section, run backwards, is the mathematics of the navigator's chart.

Try this next

1. **Solving equations.** Find all complex w with $e^w = -1$, and all w with $e^w = 1 + \mathrm{i}$.
2. **The image of a square.** Find and sketch the image under $z = e^w$ of the square with corners 0, 1, $1 + \mathrm{i}$ and i.
3. **Right angles.** Show directly that the circle $|z| = e^c$ and the ray $\arg z = d$ meet at right angles, and explain why this is consistent with the lines $u = c$ and $v = d$ being perpendicular.
4. **Squaring.** Let c and d be non-zero real numbers. Find the images of the lines $x = c$ and $y = d$ under the map $z \mapsto z^2$. Show that they are parabolas, and that they meet at right angles wherever they meet. (What happens if $c = 0$ or $d = 0$?)

Further reading

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapters 2 and 4 treat the exponential, the logarithm and conformal maps with superb pictures.
- **Hilary Priestley, *Introduction to Complex Analysis*** (Oxford University Press). A clear first undergraduate text.
- **Ian Stewart & David Tall, *Complex Analysis*** (Cambridge University Press). Another excellent undergraduate introduction, with an emphasis on geometry.

The full guided solution is in Section 5.7:
https://beyondstep.keithmcnulty.org/ch05/complex-exponential.html